

Yuwen HUANG

Curriculum Vitae

July 2026

Thrust of Data Science and Analytics, Information Hub
The Hong Kong University of Science and Technology (Guangzhou)
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Education

- 2018–2024 **PhD in Information Engineering**
Department of Information Engineering
The Chinese University of Hong Kong (CUHK), Hong Kong
Thesis supervisor: [Prof. Pascal O. Vontobel](#) [[PhD Thesis](#)]
- 2014–2018 **Bachelor of Engineering**
School of Electronic and Information Engineering
South China University of Technology (SCUT), Guangzhou, China
Thesis supervisor: [Prof. Yuan Liu](#)

Employment

- 05/2026–
Present **Assistant Professor**
Thrust of Data Science and Analytics, Information Hub
The Hong Kong University of Science and Technology (Guangzhou), China
- 11/2024–
05/2026 **Postdoctoral Fellow**
Department of Computer Science and Engineering
The Chinese University of Hong Kong (CUHK), Hong Kong
Supervisor: [Prof. John C.S. Lui](#)

Research Interests

Quantum networks, quantum information processing, probabilistic graphical models, optimization, information theory

Research Grants

- 10/2026–
09/2028 **Throughput guarantees for quantum computing networks via hop-by-hop error correction**
Young Talent Development Program, Guangzhou Municipal Education Bureau
Principal Investigator, RMB 50,000
- 10/2026–
10/2028 **Provable belief-propagation decoding of quantum low-density parity-check codes for fault-tolerant quantum computing**
Young Innovative Talents Program, Department of Education of Guangdong Province
Principal Investigator, RMB 100,000

Publications

(* indicates the corresponding author)

Journal Papers (Submitted)

- [1] ***Y. Huang** and P. O. Vontobel, “Characterizing the Bethe partition function of double-edge factor graphs via graph covers (extended version),” submitted to *IEEE Trans. on Inf. Theory*. [[arxiv](#)].
- [2] ***Y. Huang**, X. Lin, B. Luo, and J. C. S. Lui, “A scalable distributed quantum optimization framework via factor graph paradigm,” submitted to *Quantum*. [[arxiv](#)].

- [3] ***Y. Huang** and P. O. Vontobel, "The Bethe partition function and the SPA for factor graphs based on homogeneous real stable polynomials (extended version)," submitted to *IEEE Trans. on Inf. Theory*.
- [4] ***Y. Huang** and P. O. Vontobel, "Sets of marginals and Pearson-correlation-based CHSH inequalities for a two-qubit system (extended version)," submitted to *IEEE Trans. on Inf. Theory*.

Journal Papers (Published)

- [5] ***Y. Huang**, N. Kashyap and P. O. Vontobel, "Degree-M Bethe and Sinkhorn permanent based bounds on the permanent of a non-negative matrix," *IEEE Trans. on Inf. Theory*, vol. 70, no. 4, pp. 5289–5308, Jul. 2024. [[ieeexplore](#)].
- [6] **Y. Huang**, ***Y. Liu**, and F. Chen, "NOMA-aided mobile edge computing via user cooperation," *IEEE Trans. on Commun.*, vol. 68, no. 4, pp. 2221–2235, Apr. 2020. [[ieeexplore](#)].
- [7] **Y. Huang**, ***Y. Liu**, and G. Y. Li, "Energy efficiency of distributed antenna systems with wireless power transfer," *IEEE J. Sel. Areas Commun.*, vol. 37, no. 1, pp. 89–99, Jan. 2019. [[ieeexplore](#)].
- [8] **Y. Huang**, M. Liu, and ***Y. Liu**, "Energy-efficient SWIPT in IoT distributed antenna systems," *IEEE Internet Things J.*, vol. 5, no. 4, pp. 2646–2656, Aug. 2018. [[ieeexplore](#)].

Conference Papers (Published)

- [9] **Y. Huang** and ***X. Pan**, "PLASH: provably linear-time attention with selective higher-order feature sketching," in *Proc. Forty-third International Conference on Machine Learning (ICML)*, Seoul, South Korea, Jul. 2026, pp. 1–44. [[link](#)].
- [10] B. Luo, ***Y. Huang**, J. Allcock, X. Lin, S. Zhang, J. C.S. Lui, "Quantum algorithms for finite-horizon Markov decision processes," in *Proc. Forty-second International Conference on Machine Learning (ICML)*, Vancouver, Canada, Jul. 2025, pp. 41200–41234. [[link](#)].
- [11] ***Y. Huang** and P. O. Vontobel, "The Bethe partition function and the SPA for factor graphs based on homogeneous real stable polynomials," in *Proc. IEEE Int. Symp. Inf. Theory*, Athens, Greece, Jul. 7–12, 2024, pp. 3029–3034. [[ieeexplore](#)].
- [12] ***Y. Huang** and P. O. Vontobel, "Bounding the permanent of a non-negative matrix via its degree-M Bethe and Sinkhorn permanents," in *Proc. IEEE Int. Symp. Inf. Theory*, Taipei, Taiwan, Jun. 25–30, 2023, pp. 2774–2779. [[ieeexplore](#)].
- [13] ***Y. Huang** and P. O. Vontobel, "On the relationship between the global minimum of the Bethe free energy function of a factor graph and sum-product algorithm fixed points," in *Proc. IEEE Inf. Theory Workshop*, Mumbai, India, Nov. 6–9, 2022, pp. 666–671. [[ieeexplore](#)].
- [14] ***Y. Huang** and P. O. Vontobel, "Sets of marginals and Pearson-correlation-based CHSH inequalities for a two-qubit system," in *Proc. IEEE Int. Symp. Inf. Theory*, Melbourne, Australia, Jul. 12–20, 2021, pp. 1338–1343. [[ieeexplore](#)].
- [15] ***Y. Huang** and P. O. Vontobel, "Characterizing the Bethe partition function of double-edge factor graphs via graph covers," in *Proc. IEEE Int. Symp. Inf. Theory*, Los Angeles, CA, USA, Jun. 21–26, 2020, pp. 1331–1336. [[ieeexplore](#)].
- [16] **Y. Huang** and ***Y. Liu**, "User cooperation for NOMA-based mobile edge computing," in *Proc. IEEE Int. Conf. Commun. Syst.*, Chengdu, China, Dec. 19–21, 2018, pp. 395–400. [[ieeexplore](#)].
- [17] **Y. Huang** and ***Y. Liu**, "Energy-efficient SWIPT in distributed antenna systems," in *Proc. IEEE Globecom Workshops*, Singapore, Dec. 4–8, 2017, pp. 1–6. [[ieeexplore](#)].

Talks

Dec. 2025 **From Constrained Counting to Quantum Decisions: Graphical Models and Quantum Algorithms for Future Information Systems**

Seminar talk in the 6th SJTU SPEIT International Scholars Forum at the SJTU Paris Elite Institute of Technology (SPEIT), Shanghai Jiao Tong University, China

- Feb. 2025 **The Bethe approximation for binary contingency table counting and nonnegative matrix permanents**
Seminar talk (via Zoom) in the Department of Computer Science at TU Dortmund University, Germany
- Sept. 2024 **The Bethe partition function and the SPA for factor graphs based on homogeneous real stable polynomials**
Seminar talk (via Zoom) in the School of Cyber Science and Engineering at Southeast University, China
- Mar. 2023 **Bounding the permanent of a non-negative matrix via its degree- M Bethe and Sinkhorn permanents**
Seminar talk in the Institute of Theoretical Computer Science and Communications at CUHK
- Nov. 2022 **On the relationship between the minimum of the Bethe free energy function of a factor graph and sum-product algorithm fixed points**
Seminar talk in the Institute of Theoretical Computer Science and Communications at CUHK
- Apr. 2022 **Sets of marginals and Pearson-correlation-based CHSH inequalities for a two-qubit system**
Seminar talk (via Zoom) for Prof. Marco Tomamichel's research group at the National University of Singapore

Student Supervision

Research Assistants

- 2026–Present **Ms. Maoli LIU**, PhD graduate, Department of Computer Science and Engineering, The Chinese University of Hong Kong
- 2026–Present **Mr. Chun Hei SHIU**, PhD student (Year 2), Department of Electrical and Computer Engineering, University of British Columbia
- 2026–Present **Mr. Zeyu ZHANG**, PhD student (Year 2), Department of Computer Science and Engineering, The Chinese University of Hong Kong
- 2026–Present **Ms. Yu JIANG**, PhD student (Year 1), Department of Computer Science and Engineering, The Chinese University of Hong Kong
- 2026–Present **Mr. Deyun ZHANG**, Master's graduate, Department of Mathematics, Southern University of Science and Technology
- 2026–Present **Mr. Yingxian HUI**, Undergraduate student, Department of Mathematics, The Chinese University of Hong Kong (Shenzhen)

Professional Service

- Reviewer for **Journals**: IEEE Communications Magazine, IEEE Transactions on Communications (TCOM), IEEE Journal of Selected Topics in Signal Processing (J-STSP), IEEE Communications Letters
- Reviewer for **Conferences**: IEEE International Symposium on Information Theory (ISIT), International Conference on Machine Learning (ICML)

Awards

- 2024 **Student Travel Grant**, 2024 IEEE International Symposium on Information Theory (ISIT)
- 2018 **Bachelor's Thesis Award**, South China University of Technology